



PAPER ID-310484

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Subject Code: MTCS101

Roll No: 2500020105012

MTECH
(SEM I) THEORY EXAMINATION 2025-26
FOUNDATION OF COMPUTER SCIENCE

TIME: 3 HRS

M.MARKS: 70

4. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question																																																																					
a.	Explain the Banker's Algorithm for Deadlock Avoidance with Consider the following record of a system: <table><tr><th rowspan="2">Process</th><th colspan="3">Allocated</th><th colspan="3">Maximum</th><th colspan="3">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th></tr><tr><td>P1</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>P2</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P3</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P4</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P5</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td></tr></table> Answer the following questions using the banker's algorithm: 1) What is the content of the matrix need? 2) Is the system in a safe state?	Process	Allocated			Maximum			Available			A	B	C	A	B	C	A	B	C	P1	0	1	0	7	5	3	3	3	2	P2	2	0	0	3	2	2				P3	3	0	2	9	0	2				P4	2	1	1	2	2	2				P5	0	0	2	4	3	3			
Process	Allocated			Maximum			Available																																																															
	A	B	C	A	B	C	A	B	C																																																													
P1	0	1	0	7	5	3	3	3	2																																																													
P2	2	0	0	3	2	2																																																																
P3	3	0	2	9	0	2																																																																
P4	2	1	1	2	2	2																																																																
P5	0	0	2	4	3	3																																																																
b.	Explain Virtual Memory. Discuss the concept of Demand Paging and Page Fault handling.																																																																					

5. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question
a.	Explain Pushdown Automata and its relation to Context-Free Languages.
b.	Discuss the P vs NP problem. Define NP-Complete and NP-Hard problems with examples.

6. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question
a.	Explain Deadlock prevention and detection techniques in Distributed Databases.
b.	Describe the Architecture of a Distributed Database System. What are the concepts of Fragmentation and Replication?

7. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question
a.	Discuss Backup and Recovery concepts. Explain Immediate Update and Deferred Update techniques
b.	Explain Quick Sort and compare it with Merge Sort.